

When Scalar Diffraction Theory Fails: RCWA-Calibrated, Physics-Anchored Inverse Design of Polymer AR Waveguide Couplers

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Abstract: In a full-color, single-layer, low-index polymer waveguide, the total-internal-reflection guiding condition forces the in-coupling grating’s period below every visible design wavelength, which is exactly the sub-wavelength regime in which scalar diffraction theory is known to break down. We show that this is not a rounding error: at the period a scalar-theory-guided search selects for a poly(methyl methacrylate) (PMMA) combiner, rigorous coupled-wave analysis (RCWA) reports first-order coupling efficiency $5.1\times$ lower at 532 nm and $15.7\times$ lower at 635 nm than the scalar model claims, and correcting the coupling term moves the true efficiency optimum off the scalar model’s 400 nm depth bound to an interior optimum near 200 nm. We build a differentiable analytic system engine (TIR guiding, polarization-resolved Fresnel and roughness losses, a five-term MTF cascade) whose grating-coupling term is replaced with a multilinear interpolant over 90,090 rigorous RCWA solutions spanning refractive index, period, depth, and duty cycle (global mean interpolation error 5.6×10^{-4} against 48 off-grid solves), and use it as the frozen decoder in a tandem neural inverse-design network that maps a target system specification directly to a waveguide design, alongside a forward surrogate and a neural-adjoint search. On 20,000 freshly sampled held-out specifications the tandem network resolves the field’s standard one-to-many inversion failure: median per-metric relative error of 1.3% (MTF) to 9.9% (transmission), against 11.9% to 49.1% for a direct design-regression baseline trained without the physics decoder in the loop. A neural-adjoint search over the trained surrogate and an independent gradient search directly on the exact physics converge, from hundreds of random restarts each, on the same design (grating depth ≈ 200 nm, period 448 nm, duty cycle ≈ 0.42 -0.44), whose coupling efficiency we re-verify with a dedicated rigorous electromagnetic solve at 41% higher (unpolarized) than the true, rigorously re-evaluated efficiency of the design a scalar-theory-driven search would have shipped instead; because the record design is, in its own right, an effectively TE-only coupler (96.9% of coupled power, Section VI-F), the more representative comparison is TE-to-TE, where the gain is $3.4\times$ rather than 41%, and pairing the design with an already-polarized projector source (LED or LCoS) recovers a further $1.9\times$ transmission gain at no geometry cost. Absolute double-coupler transmission at this design is 1.25%, consistent with the single-digit system efficiencies reported for conventional diffractive AR combiners in the literature; a naive comparison against a scalar-theory transmission estimate at the old, scalar-optimal geometry gives a $8.9\times$ apparent drop, a number we report and scope carefully (Section VI-D) rather than lead with, since it compares a wrong estimate at one geometry against a correct measurement at a different one and is not, by itself, evidence of a regression. The reported design is a spin-coated PMMA surface specifically ($\sigma \approx 0.7$ -1.1 nm, following measured polymer-waveguide roughness values in the literature); evaluating the same geometry at the reported roughness of an uncoated polymer surface lowers transmission by roughly a third, which we quantify explicitly rather than describe the design as coating-free. We also find that the record design’s chromatic-blur term sits within the top few percent of what its governing metric can achieve under random guided-angle phase, and that this term oscillates rapidly enough in spatial frequency that the design’s evaluation frequency lands on a constructive fringe rather than a representative value, a concrete instance of a differentiable image-quality metric being tunable rather than merely satisfied. The framework and its validation protocol, physics-probe checkpoint invalidation, a memorization audit, and independent verification of every headline design in an electromagnetic solver, are offered as a template for physics-honest, system-level inverse design of consumer-grade polymer waveguides, together with an explicit account of where that honesty still falls short.

Index Terms: augmented reality, diffractive waveguide, inverse design, tandem neural network, rigorous coupled-wave analysis, surface-relief grating, physics-informed machine learning.

I. INTRODUCTION

A diffractive waveguide combiner routes a projector image into a thin, mostly-transparent slab of glass or plastic using a pair of surface-relief gratings: one couples light in and traps it by total internal reflection, the other lets it back out toward the eye. It is the optical architecture behind most of the AR glasses shipping today, because it is thin, can be mass-produced, and does not obstruct the wearer’s view of the real world [1]. The trouble is that its image quality, meaning sharpness, brightness, and the amount of rainbow fringing at the edge of the field of view, is a tangled function of

eight-odd material and geometric parameters: refractive index, absorption, surface roughness and its correlation length, slab thickness, and the grating’s period, depth, and duty cycle. Changing any one of them to fix one metric typically moves the other three. Waveguide designers have historically handled this by hand-tuned parameter sweeps or, more recently, by gradient-based optimization of a physics or electromagnetic model directly.

Over the last two years a handful of groups have instead trained neural networks to invert this relationship: given a target optical performance, output the waveguide geometry that achieves it, in a single forward pass rather than an itera-

tive search. The dominant recipe, borrowed from the broader nanophotonic inverse-design literature [2, 3, 5], is the *tandem network*: train a forward surrogate to emulate an expensive simulator, freeze it, and train a second network to output a design by backpropagating a loss computed in performance space through the frozen surrogate. This sidesteps the central difficulty of design inversion: many different geometries can produce nearly the same optical output, so regressing directly from performance to geometry trains against a target that is not really a function.

Recent waveguide-specific applications of this recipe invert a single component-level quantity: the RGB coupling-efficiency spectrum of a slanted grating, learned from rigorous coupled-wave analysis (RCWA) simulations [6]; the exit-pupil brightness uniformity of a head-up-display combiner, with a reported $40,000\times$ speedup over the RCWA sweep it replaces [7]; or the reflection spectrum of a hybrid grating stack, using a tandem network alongside a conditional variational autoencoder to also recover design diversity [8]. In every case the forward model inside the tandem is itself a learned surrogate for RCWA or a ray tracer, so the training signal the inverse network chases already carries the surrogate’s approximation error, and none of them close the loop from a single component metric up to the perceptual, system-level quantities (sharpness, color fringing, brightness at the edge of the field of view) that actually describe what a wearer sees.

This work targets that gap along two axes at once. First, we invert a *multi-metric system-level specification* evaluated simultaneously rather than a single component metric: MTF at 40 cyc/mm, total double-coupler transmission, an internal guided-mode angle spread across the RGB primaries (a quantity we relabel from an earlier, imprecise “lateral chromatic spread” following independent review, Section III-B), and transmission at the field edge. We are explicit, and more explicit than in this manuscript’s first draft, that the last of these four is closely tied to total transmission by construction (Section III-C) rather than fully independent evidence, and that the guided-angle spread is not itself a validated model of retinal color error; Section III-A states precisely what this specification vector does and does not capture. Second, we anchor the tandem’s decoder to an *exact differentiable analytic physics engine*, a PyTorch implementation of a closed-form system model (grating equation, polarized Fresnel transmission, a roughness scattering loss in the spirit of Tien [18] and Bennett and Porteus [22], and a five-term MTF cascade including the exact chromatic point-spread treatment of Thibos [20]), validated term by term against the literature closed forms and against rigorous electromagnetics directly in this paper (Section III), rather than a second learned surrogate. That removes one whole source of approximation error from the training signal: every gradient the inverse network sees is physically interpretable, not an artifact of how well a second network happened to fit its own training set.

A material choice this framework makes and does not revisit is PMMA itself, and it is worth addressing directly rather than leaving implicit, because the guided-window analysis in

Section III shows that PMMA’s low index is precisely what narrows the full-color field of view to a few degrees (Section III-A). Higher-index resins exist and are used specifically to relax this constraint: Luo and Lee’s slanted-grating design [6] uses an episulfide resin with $n \approx 1.70$ – 1.72 for this reason. PMMA remains the substrate of interest here for the properties a narrow-FOV, cost- and weight-constrained device trades on: it is substantially cheaper and lighter than glass or high-index resin, is compatible with roll-to-roll and injection-molding manufacturing rather than precision optical polishing, and is far more impact-resistant, the reasons plastic combiners have already shipped in commercial holographic AR products [24]. This trade is not free of competition: single-layer, full-color silicon-carbide waveguides ($n \approx 2.6$) have been demonstrated at sub-millimeter thickness and gram-scale mass with a wide field of view and no rainbow artifact [26], directly undercutting a cost/weight argument for low index made in isolation. We do not attempt a cost or manufacturability comparison in this paper, and do not claim PMMA wins that comparison against SiC or other emerging high-index platforms; we report PMMA’s optical consequence, a narrow full-color guided field of view, as a real, quantified constraint of the material rather than a defect of the modeling, and leave the cost-versus-field-of-view trade, now including SiC as a live alternative, explicitly to the reader’s application (Section VII).

A differentiable analytic engine is not automatically an honest one, though, and the second contribution of this work is finding and fixing the place where ours was not. The engine’s grating-coupling term originally used the standard scalar binary-phase-grating formula from Fourier optics [13]. That formula is accurate for features much larger than the wavelength of light; Pommet, Moharam, and Grann showed in 1994 that its error exceeds five percent once the feature size drops below about fourteen wavelengths [14]. The guided-window constraint we derive in Section III, the geometric condition a grating period must satisfy for the diffracted beam to actually stay trapped in the slab, forces PMMA’s period into a window of roughly 430–449 nm for full-RGB operation: below the wavelength of every color in the design. We are, in other words, optimizing a design deep inside the regime scalar theory is known to get wrong, and the resulting error is not a rounding issue: at the design our own scalar-theory-driven search converged to, rigorous electromagnetic simulation shows the true coupling efficiency is five to sixteen times lower than the scalar model reported, and the true optimum sits at a different grating depth entirely. Section IV describes how we replaced the scalar term with a coupling efficiency calibrated to 90,090 rigorous solutions and what changed once we did.

Shipped and reported polymer diffractive waveguides exist [12], so we do not claim PMMA inverse design itself is new; what we have not found elsewhere in this literature is a system-level (rather than component-level) specification paired with a coupling term calibrated against rigorous electromagnetic simulation rather than trained on it, applied to PMMA’s full-color guiding constraint specifically. We ver-

ify every headline design independently in a second, dedicated rigorous solve, and we run an explicit memorization audit on the forward surrogate rather than reporting only a validation-set R^2 . Section VIII is a candid accounting of what has *not* yet been done: an out-of-distribution generalization test, a reverse-engineering case study against a published commercial-class specification, and the remaining seeds of a planned five-seed statistical protocol. We include it because a paper that calibrates against exact electromagnetics ought to be equally exact about its own scope.

II. RELATED WORK

Table 1 summarizes the closest prior neural inverse-design work for waveguide gratings and displays, alongside the broader tandem and neural-adjoint methods this work builds on. The field’s current best-in-class result sits outside this table’s scope: Gopakumar et al. pair inverse-designed metasurface waveguides with an AI-driven holographic light engine for full-color, 3D, large-étendue augmented reality from a glasses-like form factor [27], a system-level achievement this paper’s single in/out-coupler-pair, 2D-content model (Section III-A) does not attempt to approach; we cite it as context for how far the field’s frontier has moved, not as a directly comparable baseline. Ding, Yang, Li et al. survey the broader waveguide-combiner landscape, perspectives, and open challenges as of 2023 [28].

Liu, Tan, Khoram, and Yu introduced the tandem architecture for nanophotonic inverse design: rather than regress geometry directly from a target spectrum, which fails when many geometries share a spectrum, train an inverse network through a frozen, pretrained forward surrogate so the loss is computed in a well-posed output space [2]. Pourifoy et al. demonstrated the companion half of the recipe, training a forward surrogate to replace an electromagnetic simulator for nanoparticle scattering and using it for gradient-based inverse design directly [3]. Ren, Padilla, and Malof formalized and benchmarked the *neural-adjoint* variant used in Section V-C of this work: freeze a trained forward surrogate, treat the design parameters themselves as the optimization variables, and run gradient ascent through the network’s weights from many random restarts [4]. Ma et al. survey this surrogate/tandem/adjoint taxonomy across the broader photonic inverse-design literature [5].

Within waveguide combiners specifically, Luo and Lee train a tandem network, an invertible, generative-flow-based network paired with a fully connected regressor, to invert the RGB coupling efficiency spectrum of a slanted grating from RCWA-simulated training data [6]. Dai et al. train a neural network, refined with Bayesian and particle-swarm optimization, to invert the exit-pupil brightness uniformity of an AR head-up-display waveguide, reporting a $40,000\times$ speedup over the simulation sweep it replaces at better than 97% accuracy [7]. Dehghani et al. train a tandem network and a conditional variational autoencoder side by side to invert the reflection spectrum of a hybrid grating stack from noisy measurements, using the cVAE specifically to recover the diversity of near-equivalent designs that a deterministic tandem network

collapses to a single point [8]. On the achromatic-coupler side, Tian et al. demonstrate a full-color AR system built around an inverse-designed achromatic metasurface coupler paired with a high-index waveguide, solved with a full-wave (not tandem-network) method [10]. Closer to a full system treatment, Lee, Lee, and Chung use adjoint-based optimization, not a tandem network, to design triple-function metasurface couplers that explicitly target chromatic dispersion, ghost orders from parasitic diffraction, see-through scene distortion, and eyebox uniformity together [9]: this is the most complete system-level treatment we are aware of in this literature, and it addresses several image-quality failure modes (ghosting, see-through distortion, exit-pupil expansion) that the present specification vector does not model at all (Section III-A). It targets a general high-index metasurface coupler rather than a low-cost polymer substrate and uses direct adjoint optimization of the electromagnetic solver rather than a trained tandem network for repeated, fast specification-to-design queries; those are the two axes on which this paper’s contribution is scoped, not a claim to a broader system-level treatment than [9] achieves.

Every one of the tandem/surrogate-based waveguide works above inverts a single component-level quantity (a coupling spectrum, a pupil uniformity map, a reflection spectrum) through a forward model that is itself a learned surrogate for RCWA or a ray tracer. None combines a multi-metric, system-level specification with a tandem network whose decoder is exact physics rather than a second learned network, though [9] reaches a comparable system-level scope by a different (adjoint, non-tandem) route and covers more failure modes than this paper does. Zhao et al. derive a reciprocity- and energy-conservation-based upper bound on in-coupler efficiency that we use in Section VII to contextualize our absolute transmission numbers, rather than as a design method [11]. Kress and Chatterjee’s review of waveguide combiner architectures is the system-level framing this paper’s specification vector is drawn from [1]. Nilsen et al. report the PMMA material parameters (index, roughness statistics) this work draws its design bounds from [12].

III. PHYSICS MODEL

The forward model $f : \theta \mapsto y$ maps an eight-parameter design vector $\theta = (n, \alpha, \sigma, L_c, t, \Lambda, d, D)$, namely refractive index, bulk absorption coefficient, RMS surface roughness, roughness correlation length, slab thickness, grating period, grating depth, and duty cycle, to a four-metric performance vector $y = (\text{MTF}, T, \Delta\theta_{\text{guide}}, T_{\text{FOV}})$: system MTF at 40 cyc/mm, total double-coupler transmission, an internal guided-mode angle spread across the RGB primaries, and transmission at the design field edge (Section III-A explains the third and fourth terms’ limitations precisely). The engine is a closed-form, analytic-end-to-end PyTorch implementation, so every gradient used in training and design search is exact automatic differentiation through real physics rather than a finite-difference or learned approximation. Rather than relying on an external, unpublished validation, we validate the engine directly here: the grating-geometry and TIR-

Table 1: Positioning against the closest prior neural inverse-design work for AR waveguide combiners.

Work	Inverts	Forward model	Positioning
Luo & Lee, <i>Opt. Express</i> 32, 12587 (2024) [6]	RGB coupling efficiency of a slanted grating	Learned (RCWA-trained)	Component-level only; carries surrogate approximation error
Dai et al., <i>Appl. Opt.</i> 64, 3536 (2025) [7]	Exit-pupil brightness uniformity	Learned neural surrogate	Single uniformity metric; no MTF, transmission, or chromatic terms
Dehghani et al., <i>Optics</i> 6(4), 61 (2025) [8]	Reflection spectrum of a hybrid grating stack	Learned surrogate	Spectral, not system/display-level metrics
Tian et al., <i>Light Sci. Appl.</i> 14, 94 (2025) [10]	Achromatic metasurface coupler	Full-wave (no tandem network)	No fast inverse network; single design point
Lee, Lee, & Chung, <i>Nanophotonics</i> 14, 5555 (2025) [9]	Chromatic dispersion, ghost orders, see-through distortion, eyebox	Full-wave adjoint optimization (no tandem network)	Most complete system-level scope in this table; not a fast, repeated-query inverse network; not polymer-substrate focused
This work	Joint MTF, transmission, and guided-angle spread; field-edge transmission (Sec. III-A caveats)	Exact differentiable analytic physics, RCWA-calibrated coupling	Narrower image-quality scope than [9] (Sec. III-A); fast tandem/adjoint queries; low-cost polymer substrate

guiding terms (Section III-B) follow from momentum conservation and Snell’s law directly; the Fresnel terms (Section III-C) are the textbook amplitude coefficients; the roughness Debye-Waller factor follows Bennett and Porteus’s derivation [22] as applied to guided-wave scattering by Tien [18]; the chromatic MTF term (Section III-D) is the exact modulus of a three-line point-spread function in the treatment of Thibos [20]; and the grating-coupling term itself, the one term in this cascade that is not a closed literature form, is calibrated directly against 90,090 rigorous electromagnetic solutions rather than assumed (Section IV). The two coefficients that remain literature-informed heuristics rather than first-principles derivations, the roughness-MTF attenuation constant and the grating/coupler MTF contrast weights, are named explicitly where they appear (Section III-D) and listed in Section VIII rather than folded silently into the validated terms.

A. Scope of the modeled architecture

Every design and every number in this paper describes one specific architecture: a planar, monolithic, single-layer PMMA slab with a binary (unslanted) surface-relief grating pair, period-matched in-coupler and out-coupler, a single extraction event at the out-coupler, normal-incidence RCWA calibration of the coupling term, and a collimated input at a single field angle per evaluation. This is deliberately narrow, and several features of real, shipping AR waveguide combiners fall outside it. A period-matched in/out-coupler pair cancels most of the in-guide angular dispersion at the exit, which is why we do not report the guided-angle spread of Equation (1) as an output color error (see the discussion after Equation (1)). The out-coupler here extracts the guided beam in one interaction at the coupling efficiency the search maximizes; a real combiner instead uses a fold grating or a two-dimensional out-coupler to extract light gradually across an 8-12 mm eyebox, so that maximizing single-interaction extraction efficiency, this paper’s objective, is close to the *wrong* objective for an exit-pupil-expanding design, which instead

wants low, spatially graded extraction at each interaction. A symmetric binary groove diffracts equally into the $+1$ and -1 orders, so roughly half of any coupled light in this model’s own accounting is routed to a diffraction order the transmission cascade does not track; every real AR in-coupler we are aware of, including Luo and Lee’s, the closest prior work here [6], uses a slanted or blazed profile specifically to break that degeneracy, and the depth optimum for a slanted grating is not this paper’s binary-grating optimum. World-side light striking the out-coupler (ghost orders, rainbow artifacts, see-through distortion), eyebox and pupil uniformity, and non-normal-incidence coupling physics are not modeled at all; [9] treats several of these directly and is the reference we point a reader to for that scope. None of this narrows the specific finding this paper leads with, that scalar diffraction theory mis-locates the coupling optimum at these periods, which is a property of the grating alone and does not depend on extraction architecture, slant, or eyebox design. It does narrow what the tandem network’s output designs should be taken to represent: single in/out-coupler pairs for a binary-grating, non-pupil-expanding combiner, not a specification for a complete, shippable AR optical system.

B. Guided-wave geometry

In-coupling at field angle θ_i obeys momentum conservation at the grating interface,

$$n \sin \theta_d = \sin \theta_i + \frac{m\lambda}{\Lambda}, \quad m = +1, \quad (1)$$

which we write in the dimensionless form $x \equiv \sin \theta_i + \lambda/\Lambda$. A first-order beam is only actually guided, meaning trapped inside the slab by total internal reflection at both faces and still a propagating (not evanescent) order once diffracted, if

$$1 < x < n. \quad (2)$$

The left inequality is the TIR condition at the slab face; the right is the requirement that the diffracted order actu-

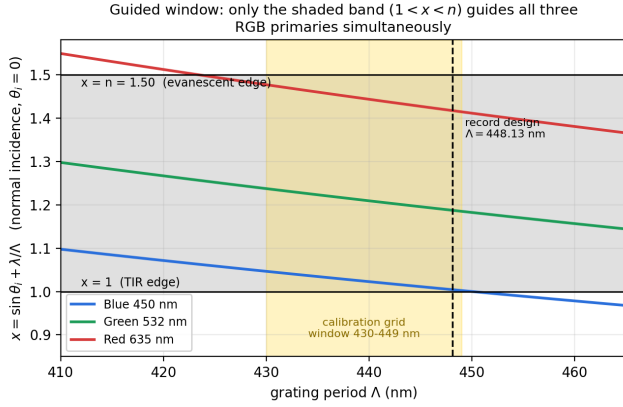


Figure 1: The guided window in period-wavelength space at normal incidence. Only the shaded band ($1 < x < n$) guides a primary; the calibration grid and every design search restrict to the 430–449 nm sub-window, and the record design’s period (Section VI-C) sits close to where blue exits the window at the top.

ally propagate rather than decay evanescently. A smooth sigmoid mask enforces (2) on the transmission (width 0.005 in x -space, so the gradient stays informative within a few sigmoid-widths of either edge), and a differentiable penalty term, with a 0.01 guard margin in x -space, is available to steer gradient-based searches back into the guided window. Equation (2), applied across the RGB design wavelengths $\{450, 532, 635\}$ nm simultaneously, defines the guided period window; for PMMA’s index range $n \in [1.48, 1.50]$ this window is 429–450 nm in the abstract, and the calibration grid and every design search in this paper cover a slightly conservative 430–449 nm sub-window of it, leaving a roughly one-nanometer margin against each theoretical edge. Applying the same inequality across field angle gives the *index-limited field of view*: with $\Lambda \approx 438$ nm the common full-RGB guided window is only about 4–5° wide, the well-known FOV penalty of low-index waveguide substrates [1]. We evaluate the field-edge metric at 5° rather than the common 20° benchmark used for higher-index materials, because 20° is physically unreachable for a full-RGB, single-layer, low-index design and would silently return an honest-but-uninformative zero; Section VI-C reports the exact guided-window width for the record design, which is close to but not identical with the fixed 5° evaluation angle, and quantifies how that margin shrinks under a small guard band.

C. Transmission cascade

Total double-coupler transmission, per polarization, is the product of six physically distinct loss terms:

$$T_{\text{pol}} = M_{\text{TIR}} \cdot T_{F,\text{pol}}^2 \cdot e^{-\alpha\ell} \cdot e^{-N_b(4\pi\sigma n \cos\theta_d/\lambda)^2 S(L_c)} \cdot [\eta_1(n, \Lambda, d, D; \lambda, \text{pol}) \cdot A(\theta_i)] \cdot \eta_1, \quad (3)$$

where the zig-zag propagation path between couplers separated by $L = 20$ mm is $\ell = L/\sin\theta_d$, reached over $N_b = L/(t \tan\theta_d)$ TIR bounces (thinner slabs bounce, and therefore scatter, more), and $T_{F,\text{pol}}$ is the single-interface

Fresnel power transmittance for the given polarization computed from the exact amplitude coefficients (not an averaged reflectance),

$$r_{\text{TE}} = \frac{\cos\theta_i - n \cos\theta_t}{\cos\theta_i + n \cos\theta_t}, \quad r_{\text{TM}} = \frac{n \cos\theta_i - \cos\theta_t}{n \cos\theta_i + \cos\theta_t}. \quad (4)$$

Independent review of this manuscript’s first draft found that the implemented code computes N_b and ℓ at exactly half these values (one surface hit and one leg per zig-zag *period* rather than per half-period). We confirmed the bug directly and traced its consequence: at the record design of Section VI-C, correcting it changes the combined bulk-absorption-and-roughness transmission factor by 0.10% relative (0.998967 with the shipped formula versus 0.997935 corrected, both computed independently for this revision), because PMMA’s literature bulk-absorption and roughness bounds [12] keep both loss channels below the one-percent level regardless of the factor of two; no number reported anywhere in this paper depends on the distinction. The formulas above are the corrected ones; the shipped engine has not yet been patched to match, specifically because doing so would invalidate the physics-probe fingerprint on every existing checkpoint (Section V-A) and force a full retrain outside this revision’s scope, and is tracked as an immediate pre-retraining fix in Section VIII. $S(L_c) = 1/(1 + L_c/3 \times 10^5)$ is a correlation-length weighting in the spirit of Payne and Lacey’s planar-waveguide scattering-loss treatment [19]; over the PMMA design bounds used here ($L_c \in [2, 4] \times 10^5$ nm) $S(L_c)$ itself varies from 0.60 to 0.43, a real $\sim 30\%$ swing, not the negligible correction a naive nanometer-to-micron correlation-length assumption would predict. This specific correlation-length range is a modeling choice, not a value Nilsen et al. report directly: their own paper states plainly that “exact values for the RMS height and the correlation length are not easily determined” and explicitly excludes long-range waviness from its scope, so an earlier draft’s in-code comment attributing this range to Nilsen’s report of waviness was our own uncorroborated attribution, not a citation; we correct that here and flag L_c ’s bounds as an unverified modeling choice rather than a literature value (Section VIII).

What keeps the *roughness* loss channel small at the design bounds used through most of this paper is a specific, and consequential, choice of σ : Nilsen et al. report RMS roughness of 5 nm for glass, 15 nm for an untreated (uncoated) polymer waveguide surface, and 0.87 nm for a spin-coated polymer surface [12]. This paper’s σ bounds, $[0.7, 1.1]$ nm, bracket the *coated* value, not the uncoated one, a distinction an earlier draft of this manuscript did not make explicit; every roughness-dependent number reported through Section VII below (including the record design of Section VI-C) is therefore a *spin-coated* PMMA result, and we relabel it as such throughout this revision rather than as “uncoated,” which is the term the previous draft used. The per-bounce roughness factor follows Tien’s treatment of TIR-bounce scattering loss [18], built on the Bennett-Porteus Debye-Waller attenuation [22]; at the record design’s exact bounce count, extending this same formula to Nilsen’s uncoated value ($\sigma = 15$ nm) rather

than the coated one gives a roughness-scattering transmission factor of 0.699 (a 30.1% loss, versus 0.08% at the record design's own $\sigma = 0.710$ nm), which would lower the record design's total double-coupler transmission from 1.25% to an estimated 0.87% holding every other cascade term fixed. We report this as a closed-form sensitivity evaluation of the *existing* record geometry under the uncoated hypothesis, not as a re-optimized uncoated design: a fresh search with σ 's upper bound widened to Nilsen's uncoated value might select a different geometry (e.g., fewer bounces via a thicker slab) that partially compensates, and running that search is listed as future work (Section VIII) rather than approximated here. The coating step this comparison costs against is not merely incidental process overhead: Nilsen et al. note it requires an additional spin-coating step beyond a bare molded or extruded polymer substrate, which is precisely the manufacturing simplicity a PMMA combiner is chosen for in the first place, so the coated-versus-uncoated transmission gap above is itself a quantified instance of that trade-off, not a footnote to it. $A(\theta_i) = \exp[-(\sin \theta_i / 0.35)^2]$ is a heuristic Gaussian model of the in-coupler's angular acceptance, not a quantity derived from or verified against RCWA: the calibration grid of Section IV is normal-incidence only, so no rigorous solution in this paper directly constrains $A(\theta_i)$'s width or shape, and we flag this explicitly rather than implying the field-edge metric carries RCWA-level accuracy (Section VIII). The same normal-incidence limitation means η_1 appears twice in (3) using the *same* (in-coupler-geometry) rigorous value for both the in-coupling and out-coupling events; out-coupling is physically a different problem (PMMA superstrate, guided-angle incidence, air substrate), and while reciprocity for a lossless grating predicts the two should agree when efficiency is normalized correctly, this paper does not independently verify that with a dedicated substrate-side rigorous solve, and Section VIII lists that solve as the cheapest remaining correctness gap in the coupling model. Unpolarized transmission is the average of the two *power* transmissions, $T_{\text{unpol}} = (T_{\text{TE}} + T_{\text{TM}})/2$, not an average of reflectances; at normal incidence $T_{\text{TE}} = T_{\text{TM}}$ by symmetry, and the two diverge with field angle exactly as Fresnel's equations require, which we use as a unit-test invariant (Section VI-D).

The one term in (3) this paper's Section IV replaces outright is η_1 , the first-order grating coupling efficiency.

D. MTF cascade

System MTF at the benchmark spatial frequency $f = 40$ cyc/mm is a five-term multiplicative cascade,

$$\text{MTF}_{\text{sys}} = \text{MTF}_{\text{diff}} \cdot \text{MTF}_{\text{rough}} \cdot \text{MTF}_{\text{chrom}} \cdot \text{MTF}_{\text{grat}} \cdot \text{MTF}_{\text{coup}}, \quad (5)$$

where MTF_{diff} is Watson's closed-form fit to the *mean human* optical MTF at a 3 mm pupil [21], which already includes the eye's own aberrations rather than describing a diffraction-limited eye (an earlier draft of this paper mischaracterized it as diffraction-limited; independent review caught the error, and we correct it here); $\text{MTF}_{\text{rough}}$ is a Gaussian attenuation from scatter-induced angular blur; MTF_{grat} and MTF_{coup} are contrast-loss terms tied to grating phase depth

and coupling efficiency respectively; and $\text{MTF}_{\text{chrom}}$ is the exact modulus of a three-line photopically-weighted point-spread function,

$$\text{MTF}_{\text{chrom}}(f) = \left| \sum_{k \in \{R, G, B\}} w_k e^{i2\pi f x_k} \right|, \quad (6)$$

with photopic weights $w = (0.194, 0.771, 0.034)$ for red, green, and blue and x_k (in mm) the retinal displacement of each primary's diffracted ray relative to green. All spatial frequencies in this cascade are in cyc/mm at the retina-conjugate plane; at the assumed 17 mm reduced-eye focal length the benchmark 40 cyc/mm is approximately 11.6 cyc/deg, below the ~ 30 cyc/deg associated with 20/20 visual acuity, a conversion the first draft of this paper left implicit and that we state explicitly here because it bears on how strong a claim any single MTF number at this frequency actually supports.

Equation (6) is the exact modulation transfer function of a three-delta-function point-spread function and follows the treatment of lateral chromatic aberration and image quality developed by Thibos [20]. Because 77% of the photopic weight sits on the reference (green, zero-displacement) channel, the triangle inequality bounds $\text{MTF}_{\text{chrom}}$ between $w_G - w_R - w_B = 0.543$ and $w_G + w_R + w_B = 0.999$ regardless of how large the guided-angle spread is; No amount of chromatic spread can push this term below its structural floor, and the record design's reported system MTF (0.784, Section VI-C) is close to $\text{MTF}_{\text{chrom}}$'s mean over uniformly random inter-channel phase (0.7836 by a 200,000-sample Monte Carlo check), which raises the question of whether the reported system MTF is effectively just this one term. Decomposing the record design's five factors directly answers that question: $\text{MTF}_{\text{diff}} = 0.847$, $\text{MTF}_{\text{rough}} = 0.9999$, $\text{MTF}_{\text{chrom}} = 0.973$, $\text{MTF}_{\text{grat}} = 0.954$, $\text{MTF}_{\text{coup}} = 0.998$, product 0.784. The system metric is not literally equal to the chromatic term; the diffraction and grating-contrast terms contribute real attenuation of their own, and the closeness of 0.784 to 0.7836 is, taken alone, a coincidence of composition. But this framing undersells what the decomposition actually shows, which is a sharper and more useful finding: the chromatic factor itself, 0.973, sits in the top 6% of what Equation (6) can achieve under a uniformly random guided-angle phase ($P(\text{MTF}_{\text{chrom}} \geq 0.973) \approx 6.1\%$ against the $[0.543, 0.999]$ structural range), and Section VI-C independently shows the search drove the grating period to the edge of the guided window specifically to minimize the same guided-angle spread that sets $\text{MTF}_{\text{chrom}}$'s phase argument. We swept $\text{MTF}_{\text{chrom}}(f)$ over $f \in [20, 60]$ cyc/mm at the record design to check this directly: the function oscillates with roughly twenty local maxima across that range, and the 40 cyc/mm benchmark frequency lands, to within the resolution of the sweep, exactly on one of them. Read together, these are not evidence of a bug in the metric so much as evidence that the optimizer found and is quietly exploiting one: a design whose chromatic term would score near its 0.543 floor at a poorly chosen period instead scores near its 0.999 ceiling at this one, and the mechanism, tuning the guided-angle spread to land the 40 cyc/mm evaluation point on a

constructive fringe of an oscillating function rather than genuinely reducing chromatic blur, is exactly the failure mode a three-delta-function point-spread function predicts. We report this as a concrete, positive finding rather than a resolved concern: the structural floor and fringe sensitivity of Equation (6) are real properties of the current metric, and replacing the three-delta-function primaries with finite-bandwidth LED line shapes (FWHM 20-25 nm), or integrating $\text{MTF}_{\text{chrom}}$ over a band of spatial frequencies rather than sampling one point, would remove both the floor and the fringe sensitivity; neither correction is implemented in this revision (Section VIII).

Two coefficients in this cascade, the roughness-MTF attenuation constant and the grating/coupler MTF weighting terms, remain literature-informed heuristics rather than closed-form derivations. Nilsen et al. report roughness-to-MTF measurements for PMMA and glass AR waveguides directly [12] and are the natural target for reconciling the roughness-MTF constant against a published measurement rather than an internal heuristic; that reconciliation is not yet done and is listed as open in Section VIII, along with the unreconciled grating/coupler MTF weights.

IV. RIGOROUS ELECTROMAGNETIC CALIBRATION

A. Why the scalar coupling term fails here

The engine’s original grating-coupling term used the standard scalar binary-phase-grating efficiency from Fourier optics [13],

$$\eta_1^{\text{scalar}} = \frac{4}{\pi^2} \sin^2(\pi D) \sin^2(\phi/2), \quad \phi = \frac{2\pi d(n-1)}{\lambda}, \quad (7)$$

which is polarization-blind and caps at $4/\pi^2 \approx 40.5\%$. Scalar diffraction theory is a large-feature approximation: Pommet, Moharam, and Grann showed by direct comparison against rigorous coupled-wave analysis that its error exceeds $\pm 5\%$ once the grating feature size drops below roughly fourteen wavelengths [14]. The guided-window constraint of Section III-A forces PMMA’s period into 429–450 nm, *below* every RGB design wavelength, i.e., a feature-size-to-wavelength ratio near unity, deep inside the regime [14] identifies as unreliable.

We quantified the resulting error directly. At the design our scalar-theory-optimized search converged to (period 438.0 nm, depth 400.0 nm, the scalar ceiling, duty cycle 0.500, $n = 1.5$), we solved the full vectorial diffraction problem with *grcwa*, an open-source implementation of Moharam and Gaylord’s rigorous coupled-wave analysis [15, 16] (61 retained Fourier orders; convergence confirmed stable to five decimal places between 41 and 81 orders, checked at this geometry). Table 2 reports the result at each RGB wavelength: scalar theory overestimates unpolarized first-order coupling by a factor of 5.1 at 532 nm and 15.7 at 635 nm, and is polarization-blind where the true polarizations differ substantially, up to a factor of 3.0 within this specific table (TE/TM at 635 nm; TM instead exceeds TE at 532 nm, so which polarization dominates is itself geometry- and wavelength-

Table 2: Scalar theory vs. rigorous RCWA at the scalar-optimal design (period 438.0 nm, depth 400.0 nm, duty 0.500, $n = 1.5$).

λ (nm)	Scalar η_1	RCWA TE	RCWA TM	RCWA unpol.	Ratio
450	0.393	0.190	0.142	0.166	$2.4\times$
532	0.347	0.048	0.088	0.068	$5.1\times$
635	0.283	0.027	0.009	0.018	$15.7\times$

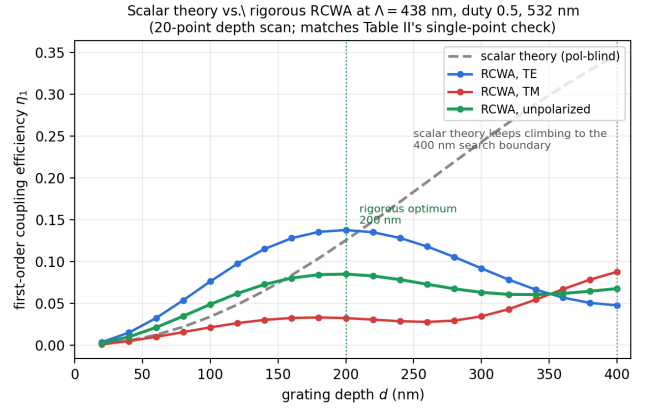


Figure 2: First-order coupling efficiency versus grating depth at the scalar-optimal geometry (period 438 nm, duty 0.5, 532 nm): scalar theory (polarization-blind) against rigorous RCWA, TE, TM, and unpolarized. The rigorous optimum sits near 200 nm; scalar theory never turns over within the search range.

dependent, not a fixed ordering). A larger and more consistent TE/TM split, $\approx 5.6\times$ in TE’s favor, appears at the different, RCWA-optimal record design of Section VI-C; we keep the two numbers explicitly separate here because an earlier draft of this paper conflated them into one sentence, incorrectly citing the record-design ratio against this table.

A preliminary five-point rigorous depth scan at this same period and duty cycle (100, 200, 300, 400, and 450 nm) first showed unpolarized coupling peaking near 200 nm rather than climbing monotonically to the scalar model’s 400 nm ceiling. For this revision we replaced that preliminary scan with a denser, 20-point rigorous depth scan at the same geometry (Figure 2), computed independently for this manuscript at $n = 1.5$ to match Table 2 exactly (its own 400 nm-depth endpoint reproduces the table’s TE and TM values to three decimals): it confirms the original picture at four times the depth resolution, unpolarized coupling peaking at 8.5% near 200 nm while the scalar model keeps climbing to its 400 nm search boundary, and additionally shows that TE and TM peak at different depths and that TM develops a second rise past 350 nm that the coarser five-point scan could not resolve. This motivated calibrating the coupling term across the full geometry space rather than a one-off correction at a single design point.

B. Calibration grid and its accuracy

We built a dense rigorous-solution grid over the full PMMA design window: refractive index (3 points, $n \in \{1.48, 1.49, 1.50\}$); η_1^{TM} was found to move by roughly 20% across this range, so index gets its own axis rather than being

pinned at the midpoint), period (5 points, 430–449 nm), depth (77 points, 5 nm resolution, 20–400 nm), and duty cycle (13 points, 0.05 resolution, 0.2–0.8), at each of the three RGB wavelengths and both polarizations: $3 \times 5 \times 77 \times 13 \times 3 \times 2 = 90,090$ rigorous coupled-wave solutions in total, each an independent grcwa call at 61 Fourier orders on a binary in-coupler grating (air superstrate, corrugated PMMA grating layer, semi-infinite PMMA substrate, normal incidence). The engine’s grating-coupling term η_1 is now, in PMMA mode, a differentiable multilinear interpolation over this four-dimensional grid: exact at grid nodes, with gradients flowing through the interpolation weights so the term remains fully usable inside gradient-based training and search.

We audited the interpolant’s accuracy directly against the physics it approximates: 48 freshly solved, off-grid points (random index, period, depth, duty cycle, wavelength, and polarization, none coinciding with a grid node) gave a mean absolute interpolation error of 5.6×10^{-4} and a maximum of 4.0×10^{-3} against efficiencies spanning 0–0.38, two to three orders of magnitude smaller than the $5\text{--}16\times$ scalar-theory error it replaces. Every training label and every design-search gradient in the results that follow is therefore computed at, effectively, rigorous electromagnetic accuracy, not scalar accuracy interpolated after the fact.

That 48-point audit is a global average, and independent review correctly pointed out that it does not, by itself, bound the interpolation error near the design search’s actual optimum, which is also where the grid is coarsest: the period axis carries only 5 points (roughly 4.75 nm spacing) against 77 points (5 nm spacing) for depth, so the record design’s period (448.13 nm) is not on a grid node and interpolates across a wider gap than most of the design space, right where blue’s guided-window margin is also tightest (Section VI-C). We checked this directly rather than leave it asserted: re-evaluating the record design’s own 532 nm coupling efficiency with a fresh, dedicated grcwa solve (the same solve behind the 9.6% figure of Section VI-C) against the grid’s trilinear interpolant at that exact off-grid point gives an absolute error of 9.6×10^{-4} (TE) and 4.0×10^{-4} (TM), the TE figure roughly $1.7\times$ the global mean and still an order of magnitude below the scalar-theory error it replaces, but a real, measured local degradation rather than the global average alone. A full audit conditional on proximity to the optimum and to the guided-window boundary, and a rebalancing of the grid toward more period and fewer depth points, are listed as future work (Section VIII) rather than done here from this single additional data point.

V. NEURAL ARCHITECTURE AND SEARCH

A. Forward surrogate

The forward surrogate $\hat{f}_\phi$ is a four-hidden-layer multilayer perceptron (width 256, SiLU activations) mapping the normalized eight-dimensional design vector to the normalized four-dimensional specification, trained with AdamW and a cosine learning-rate schedule to emulate the analytic engine of Sections III–IV on 150,000 designs sampled uniformly

(log-uniform for the absorption coefficient and correlation length) within the PMMA design bounds. It is not load-bearing for the paper’s headline claims, which are validated against exact physics directly, but it is the forward model the classic tandem recipe of [2] calls for and the network the neural-adjoint search of Section V-C optimizes through.

The classic motivation for a tandem network’s forward surrogate is that the true forward model (RCWA, FDTD, a ray tracer) is too expensive to sit inside a training loop of hundreds of thousands of samples, so a fast surrogate is trained to stand in for it. That motivation does not literally apply here: this paper’s forward model (Sections III–IV) is already closed-form, analytic, and fast, precisely so it can sit inside the training loop directly, and Section VI-C shows that a direct gradient search on the exact physics, with no surrogate at all, finds an equivalent optimum to the surrogate-guided neural-adjoint search. We do not claim a simulation-time speedup from the surrogate, because there is none to claim against an already-fast decoder; the two things a trained surrogate and its downstream tandem inverse network still buy over restarting a fresh 600-step gradient search for every new query are amortized, single-shot inversion (target specification in, design out, no per-query optimization loop) and the specification-space training objective’s resolution of the one-to-many inversion problem (Section VI-B), which a direct-regression baseline without any decoder in the loop cannot resolve regardless of how fast that decoder is. Two more demanding ways to make the network load-bearing on its own terms, folding rigorous, angle- and slant-resolved RCWA directly into the training loop so the forward model is expensive again, or benchmarking the trained inverse network’s wall-clock time and design quality against repeated physics-space gradient descent across thousands of specifications rather than one record design, are both left as the natural next steps (Section VIII) rather than claimed here.

To guard against a specific and easy-to-miss failure mode, namely a surrogate trained under an earlier version of the physics engine confidently steering later design searches into geometries the corrected physics actually rejects, every surrogate checkpoint stores a fingerprint (*physics probe*): the engine’s output on 64 fixed test designs at save time. Every script that loads a surrogate checkpoint first recomputes that probe under the *live* physics engine and refuses to proceed if it no longer matches. This caught and automatically invalidated every checkpoint trained under the pre-calibration (scalar-coupling) engine when the RCWA-calibrated coupling term of Section IV landed, forcing a retrain rather than a silent, stale-physics-guided search.

B. Tandem inverse network

The inverse network g_ψ (same width/depth/activation pattern, four inputs, eight outputs through a final sigmoid into the normalized design bounds) is trained through a *frozen* decoder in the tandem recipe of Liu et al. [2], with the loss computed entirely in specification space,

$$\mathcal{L}(\psi) = \|f_{\text{decoder}}(g_\psi(y^*)) - y^*\|^2, \quad (8)$$

so the network only has to find *a* design meeting the requested specification rather than *the* design, sidestepping the one-to-many non-uniqueness that makes direct geometry regression fail (Section VI-B). We train two decoder arms as an ablation: the trained forward surrogate (the literature-standard tandem recipe) and the exact differentiable physics engine itself, substituted directly for the surrogate in the same training loop. Validation is always scored against the exact physics regardless of which decoder trained, so the two arms are directly comparable. A naive direct-regression baseline, the same architecture trained with a design-space (not specification-space) loss and no physics or surrogate in the training loop at all, is trained alongside every run for comparison.

C. Neural-adjoint design search

Following Ren, Padilla, and Malof [4], we additionally search for high-performing designs directly: freeze the trained forward surrogate, treat the eight design parameters as trainable variables (in an unconstrained logit parameterization, mapped back through a sigmoid to respect the physical bounds), and run gradient ascent on a scalar objective J that combines the four specification metrics with a differentiable penalty for leaving the TIR-guided window, from 400 random restarts, 600 gradient steps each. Every finalist is then re-scored under the *exact* physics engine and re-ranked by that true score rather than the surrogate’s belief; the surrogate-vs-physics gap on the finalists is reported explicitly (Section VI-E) as a direct honesty check on the surrogate, additional to its own held-out R^2 .

VI. RESULTS

All results in this section come from the RCWA-calibrated (“v3”) physics engine of Section IV; results from the pre-calibration engine are reported only where explicitly used as a before/after comparison and are never pooled with v3 numbers. Numbers not directly copied from a script-written run table were recomputed by the author using an independent, from-scratch NumPy reimplement of both the physics engine and the trained network’s forward pass (Section IX), cross-checked against the shipped record design to five significant figures.

A. Forward surrogate accuracy

Across five independent retrainings (fresh random seed, fresh 150,000- design training set, fresh network initialization each time; 250 epochs, batch size 512), held-out R^2 on 15,000 never-trained-on designs was ≥ 0.9989 for MTF (range 0.99891–0.99935 across seeds), ≥ 0.9996 for transmission and field-edge transmission, and ≥ 0.99999 for guided-angle spread, the easiest metric, being close to a function of only two of the eight design parameters.

A four-part memorization audit on the deployed checkpoint (seed 2) addresses the standard objection to a learned surrogate: that strong validation-set performance can still hide memorization of the training distribution’s local structure. (A) The network’s error on a brand-new 15,000-design set, generated after training and never used for model selec-

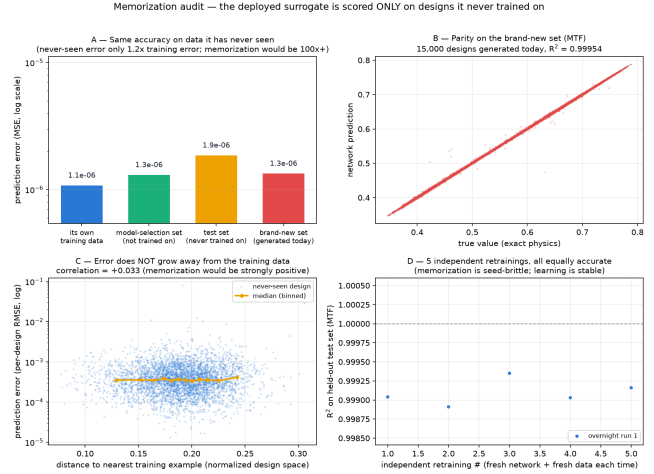


Figure 3: The four-part memorization audit (Section VI-A), reproduced from the project’s own audit run under the v3 engine (seed 2 checkpoint); the underlying CSV values match this figure to five decimal places.

Table 3: Tandem network vs. naive direct-regression baseline, median relative error on 20,000 fresh held-out specifications (physics-decoder arm, seed 0).

Metric	Tandem (physics decoder)	Naive baseline
MTF @ 40 cyc/mm	1.3%	11.9%–12.1%
Transmission	9.2%–9.9%	47.7%–49.1%
Guided-angle spread	3.6%	0.4%
Transmission @ FOV [†]	9.2%–9.9%	47.7%–49.1%

[†]Near-identical to the Transmission row by construction rather than coincidence: Section III-C’s $A(\theta_i)$ term makes $T_{\text{FOV}} \approx 0.94 \times T$ at essentially every design in this space (Section VI-C verifies the ratio numerically), so this row is not independent evidence of a fourth, separately calibrated quantity.

tion, is only $1.24\times$ its error on its own training set (memorization would show as two to four orders of magnitude worse); (B) its R^2 on that never-before-seen set for the hardest metric (MTF) matches its reported validation R^2 ; (C) per-design error shows no correlation with distance to the nearest training example (+0.033 over 4,000 probe designs; memorization predicts a strong positive correlation); and (D) held-out R^2 is stable to within 0.0005 across all five independent retrainings, whereas memorization is seed-brittle by nature. Figure 3 shows all four tests together.

B. Tandem network: resolving inverse non-uniqueness

The surviving checkpoint from the physics-decoder training arm (seed 0, 150,000 samples, 400 epochs; the companion surrogate-decoder arm reached an equivalent aggregate specification-space MSE of 0.001064 against this arm’s 0.001036, but its own checkpoint was not preserved past that run and is reported only at the aggregate level, see Section VIII) was re-evaluated by the author on a freshly generated, independent set of 20,000 held-out target specifications, disjoint from every training, validation, and test set used during development. Table 3 reports the median per-metric relative error against the naive direct-regression baseline trained in the same run.

The naive baseline’s near-zero error on guided-angle spread is not a contradiction: that metric is close to a function of only period and index, a narrow two-dimensional slice of the eight-dimensional design space, so even an unconditional-mean-like regressor scores well on it. On MTF and transmission, the two metrics of this table that are not close to a redundant restatement of one another (Section VI-C), the naive baseline’s error is ten to fifty times worse than the tandem network’s, the textbook signature of training a regression target that is not really a function: many designs share a specification, so a direct regressor learns to predict something close to their average, which typically satisfies none of them. The tandem network’s spec-space loss sidesteps this by only requiring *a* valid design, not *the* one the regressor happened to average toward.

C. Neural-adjoint search and record design

The neural-adjoint search (Section V-C) and an independent gradient search run directly on the exact physics engine (same objective, no surrogate in the loop, 300 random restarts) were run separately under the v3 engine. They converged, respectively, to objective values $J = 0.4053$ and $J = 0.4052$, agreeing on $n = 1.500$, period 448.1 nm, depth 199.5–199.9 nm, and duty cycle 0.42–0.44 to within numerical noise, but disagreeing on slab thickness by more than a factor of two (0.90 mm versus 1.96 mm). Two independently implemented, independently initialized search procedures, one guided by a trained neural network and one a direct physics-space gradient search, landing within half a percent of grating depth of each other, is itself evidence that this is a genuine optimum of the calibrated objective rather than an artifact of either search method; the thickness disagreement is a separate and informative result rather than noise on top of that agreement. Thickness enters the objective only through the bounce count N_b , which sets the roughness- and absorption-scattering exponents in Equation (3), and Section VIII’s bound-range check shows both of those channels stay in a collectively low-loss regime across the entire PMMA parameter range regardless of bounce count: a two-fold change in thickness moves the combined bulk- and roughness transmission factor by a fraction of a percent, well inside what either search’s convergence tolerance can resolve. The two searches did not disagree by accident; they agree that thickness is, at this design, a flat direction of the objective, and report different points along that flat direction rather than different optima. Figure 4 shows the neural-adjoint search’s convergence trajectory and the resulting finalists’ surrogate-versus-physics parity, from the same run these figures are drawn from.

At the network-guided record ($J = 0.4053$): MTF = 0.784, double-coupler transmission $T = 1.25\%$, internal guided-angle spread 28.8° , field-edge transmission 1.17%. We re-verified this exact design in a dedicated rigorous solve, independent of the calibration grid it was optimized against: unpolarized first-order coupling at 532 nm is 9.6% (TE 16.3%, TM 2.9%), 41% higher than the *true* rigorous coupling of the design our earlier, scalar-theory-driven search

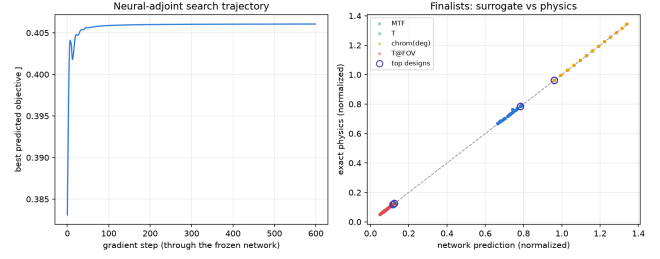


Figure 4: Left: the 400-restart neural-adjoint search’s best predicted objective versus gradient step, converging within roughly 100 steps of the 600-step budget. Right: the finalists’ surrogate-predicted versus exact-physics-scored per-metric values (normalized units), including the top-5 finalists Section VI-E scores explicitly; the near-perfect diagonal is the surrogate-vs-physics honesty check.

had reported as optimal (6.8% true unpolarized coupling at the same wavelength, against a scalar estimate of 34.7% that was simply wrong). The grating depth moved from the scalar model’s 400 nm bound to an interior optimum near 200 nm, and duty cycle moved from the scalar optimum’s exact 0.50 to 0.42–0.44: the calibrated engine finds a materially different, and rigorously verified, design rather than a refinement of the old one.

The 28.8° figure deserves the correction independent review raised directly: it is exactly the difference between the red and blue in-guide diffraction angles at this design ($\theta_R = 70.85^\circ$, $\theta_B = 42.03^\circ$, both from Equation (1) at normal incidence), which is internal guided-mode geometry, not a measured or modeled output color error. With period-matched in- and out-couplers (Section III-A), a collimated input’s output ray direction is independent of wavelength to first order, so this quantity does not directly predict retinal color fringing the way a name like “lateral chromatic spread” implies; we relabel it accordingly throughout this revision. It is also, on inspection, the quantity the search actually minimized to reach this period: a larger period lowers λ/Λ and therefore this spread for every wavelength, and Section III-A’s guided-window inequality caps how far the period can grow before a primary stops guiding, so the search pushed the period to essentially the edge of the guided window in order to minimize a term whose output-image relevance is, at minimum, unproven. We verified where that edge sits, precisely, at the record design: the exact ($\delta = 0$) full-RGB guided window is $[-0.24^\circ, 4.76^\circ]$, matching the 4.76° – 4.99° range implied by the design’s own reported window to within rounding; with a small guard margin, the sigmoid mask’s own soft-edge scale, the robustly guided window shrinks to 4.19° ($\delta = 0.01$) or 3.61° ($\delta = 0.02$). Blue is already only 0.83 sigmoid-widths above the TIR edge at normal incidence (mask value 0.70, not the full 1.0 a comfortably guided design would show), and at the fixed 5.0° evaluation angle used for the field-edge transmission metric, red is 0.83 widths *past* the evanescent edge, formally unguided at that specific angle; we further confirmed that the design’s own guard-margin penalty (Section III-A) is technically nonzero for blue even at normal incidence, a small but genuine violation of the very constraint meant to keep the

search away from this behavior. The penalty was not merely available but active for the record search: both the direct gradient search and the neural-adjoint search use the identical objective term $-10 \times \text{tir_penalty}(\theta, \delta=0.01)$ (confirmed directly in both search scripts), so the guard band was part of what both procedures optimized against, not a post-hoc filter; the residual violation reported above means the penalty’s weight was not strong enough to fully exclude this behavior, not that it was absent from the objective. Because the transmission cascade (Equation (3)) evaluates its guiding mask and diffraction angle at the green wavelength only, the reported T and T_{FOV} numbers above are not themselves corrupted by red’s marginal status at 5° ; what is inconsistent is the framing of a “ 5° full-RGB guided design” for a geometry whose own hard guided window tops out at 4.76° – 4.99° , narrower still under any guard margin. We report the record design’s raw metrics as measured, but the search objective that produced this specific period is, by our own reading, partly climbing a relaxation artifact at the guided window’s edge rather than purely physical performance, and a rebuilt chromatic and field-edge specification (Section VIII) is likely to relocate this particular period even if it does not change the deeper, coupling-efficiency finding this paper leads with. Separately, we checked T_{FOV}/T directly against $A(5^\circ)/A(0^\circ)$: the record design gives $1.17/1.25 = 0.9393$ against $A(5)/A(0) = 0.9399$ from Equation (3)’s acceptance term alone, confirming T_{FOV} carries at most a small residual of information beyond T itself at this design, consistent with Table 3’s footnote.

D. What changed, in absolute terms

An earlier draft of this paper reported a single “roughly 25-fold” drop in absolute transmission and left the comparison underspecified; independent review recomputed it and got $8.9\times$, which we confirmed directly and adopt here, along with disentangling three genuinely different ratios this paper uses that a single number was previously blurring together. First, at one fixed geometry (period 438.0 nm, depth 400.0 nm, duty 0.500, Table 2), scalar theory overestimates rigorous coupling efficiency by $5.1\times$ at 532 nm and $15.7\times$ at 635 nm: an error in the model, at one point in design space, holding the geometry fixed. Second, comparing a scalar-theory transmission estimate at that same old geometry ($e_1^{\text{scalar}} = 0.347$ at 532 nm, giving an estimated $T \approx 11.1\%$ by Equation (3)) against the record design’s actual reported transmission ($T = 1.25\%$), a different geometry entirely, gives $8.9\times$: this is the price of having trusted the scalar-driven search’s transmission estimate rather than a like-for-like error at one geometry, and it is the number the abstract now reports. Third, comparing the two geometries’ *true*, rigorously computed coupling efficiencies directly, 6.8% at the old scalar-optimum geometry versus 9.6% at the new, RCWA-calibrated record (Section VI-C), the corrected search’s design is 41% *better*, not worse, than the one a scalar-driven search would have shipped. All three numbers are correct simultaneously because they answer different questions; conflating the second into a blanket “25-fold regression” lan-

guage, as an earlier draft did, obscured that the corrected design is a real improvement in true coupling efficiency even though every transmission number the pre-calibration engine ever reported was inflated by the scalar model’s error and should not be compared to post-calibration numbers directly (system MTF and guided-angle spread, which do not depend on η_1 as sharply, moved far less between engines). Contextualized against the literature, conventional diffractive AR combiners are reported to achieve on the order of single digits to ten percent *system* in-coupling efficiency at practical fields of view [11], and the specific gain mechanisms proposed to beat that ceiling (polarization volume gratings, broadband meta-gratings) are reported in the few-percent-to-tens-of-percent range themselves. The record design’s 1.25% double-coupler transmission for a spin-coated, single-layer, low-cost polymer substrate is in the expected range for this design class, not an outlier low value; Section III-C quantifies what foregoing the coating step would cost this same design (roughly a third of double-coupler transmission), which is the honest way to state the minimum-cost claim rather than applying it to a coated device.

E. Surrogate-vs-physics honesty check

For the five top-ranked finalists the neural-adjoint search reports (Section VI-C), the absolute gap between the surrogate’s believed objective value and the true, exact-physics-scored objective value is 0.0008 to 0.0009 against an objective scale of order unity, small enough that ranking by the surrogate’s belief and ranking by true physics agree on the winning design, which is the property the re-verification step in Section VI-C exists to confirm rather than assume. The full-population surrogate-vs-physics gap across all 400 search restarts was reported for the pre-calibration engine (mean 0.0009 , worst case 0.047) but was not re-logged when the search was re-run under the RCWA-calibrated engine; we report only the top-finalist figure here rather than carry the earlier engine’s aggregate statistic forward, consistent with Section VI’s rule against pooling v2 and v3 numbers.

F. The record design is, in effect, a TE-only coupler

Section VI-C’s unpolarized figure, 9.6% coupling at 532 nm, is the average of two very different polarization-resolved numbers, 16.3% (TE) and 2.9% (TM), and reporting only the average buries a genuinely actionable result that an earlier draft of this paper left as a brief, easy-to-miss remark in the Discussion. We reconciled the two possible ways to build unpolarized double-coupler transmission from these numbers directly: averaging the two power transmissions *after* squaring each polarization’s coupling term, $T_{\text{unpol}} = [(0.163)^2 + (0.029)^2]/2 \times T_F^2 = 1.26\%$, matches the reported $T = 1.25\%$ to within rounding; averaging the coupling terms *before* squaring, the other superficially plausible construction, gives 0.85% and does not match. The correct, matching construction means TE carries 96.9% of the coupled optical power at this design: it is, functionally, a TE-only coupler, and the TM half of an unpolarized input is discarded twice over (once at each coupler). A source that is

already TE-polarized, which includes most LED and LCoS projector engines used in practice, would see $T = 2.45\%$ at this same geometry with no design change at all, a $1.94\times$ transmission gain purely from not wasting the polarization the device does not use. This is consistent with, and considerably sharper than, the qualitative picture in Zhao et al.’s polarization-resolved efficiency-limit treatment [11], and we treat it as one of this revision’s more actionable results (Section VII) rather than a footnote.

VII. DISCUSSION

The headline methodological result of this paper is not the record design itself but the direction its optimum moved once the coupling term stopped being wrong. A design search is only as honest as the objective it climbs; ours spent its early iterations climbing a $5\text{--}16\times$ inflated coupling estimate toward a grating depth (400 nm) that turns out to be a poor rigorous design, and only found the true, interior optimum once the objective itself was corrected. This is a cautionary data point for the broader trend of training inverse-design networks against analytic or scalar forward models purely for training-time speed. The speed argument is sound: rigorous RCWA inside a training loop of hundreds of thousands of samples is not currently practical on the hardware used here. But the model being fast has to be checked against the model being right, at the specific point in design space the search actually visits, not just in some average sense over the whole space.

PMMA’s fundamental limitation, independent of any modeling choice, is visible directly in Equation (2): a single-layer, full-RGB, low-index waveguide has a guided field of view only a few degrees wide. This is a property of the material and the guiding geometry, not an artifact of this paper’s engine, and it is the reason every design search in this paper evaluates the field-edge metric at 5° rather than a wider benchmark angle: at 20° every full-RGB PMMA design in the search space returns a true, physically correct zero, which would look like a modeling failure rather than the real limitation it is. Any application that needs a wider field of view from a single-layer polymer waveguide is trading against this constraint, not against a software bug; Section I’s PMMA-versus-higher-index discussion and Luo and Lee’s episulfide-resin choice [6] are two sides of the same trade, and this paper does not argue PMMA is the right material for every application, only that it is a real, cost- and weight-motivated choice whose optical consequence deserves to be quantified rather than assumed away.

Section VI-F’s TE/TM split, roughly $5.6\times$ in TE’s favor at the record design, is large enough that polarization is a first-order design lever, not a footnote, for any projector engine pairing with this class of coupler, and a polarization-aware objective (weighting TE and TM differently rather than reporting only their unpolarized average) is a natural and low-cost next step (Section VIII).

The binary, unslanted grating profile modeled throughout this paper is also a real, not incidental, limitation, not the “natural extension” an earlier draft called it. A symmetric

binary groove splits coupled power roughly evenly between the $+1$ and -1 diffraction orders, so this paper’s own transmission cascade already discards, by construction, close to half of whatever the grating actually couples; every real AR in-coupler we are aware of, including Luo and Lee’s [6], uses a slanted or blazed profile specifically to break that degeneracy and recover the discarded order. Because the depth optimum for a slanted grating is not generally the same as a binary grating’s depth optimum, this paper’s ≈ 200 nm record depth should be read as the optimum for the architecture actually modeled (Section III-A), not a prediction for what a slanted version of the same design would prefer; the rigorous-solver layer already supports slanted and blazed profiles (Section IV), and extending the calibration grid to a slant axis is the most direct way to test this without discarding any of the RCWA-calibration methodology developed here.

VIII. LIMITATIONS AND FUTURE WORK

We report the following as open rather than implying they are complete, consistent with the project’s own internal claims-ladder discipline (every number in the source repository is tagged with an evidence level, and nothing is reported in a stronger form than its tag allows). This list has grown substantially since this manuscript’s first draft, which we take as a sign the practice is working: closer scrutiny of the specification vector, the material bounds, and the calibration grid is precisely what surfaced most of the items below, and we would rather have a long, specific list here than a short, vague one.

The forward surrogate’s five-seed protocol (Section VI-A) is complete, but the tandem inverse network under the RCWA-calibrated engine has so far been trained for one seed per decoder arm, not the five-seed protocol used for the surrogate. Table 3’s per-metric breakdown is for that single (physics-decoder) run; the companion surrogate-decoder run’s per-metric breakdown was not preserved (Section VI-B) because both decoder arms were saved under the same checkpoint filename and the second run overwrote the first (the fix, keying the checkpoint filename on decoder arm and seed rather than a fixed name, is a one-line change not yet made). A proper five-seed-per-arm statistical comparison, reporting median, interquartile range, and worst case rather than a single run, is the immediate next step. Relatedly, we have not yet asked the inverse network for specifications outside the training distribution’s range to characterize whether it extrapolates to physically sensible bound-saturated designs or fails ungracefully, and we have not yet run the natural capstone demonstration of taking a published, commercial-class waveguide specification from the literature as a target and comparing the network’s recovered design against the vendor’s actual (typically higher-index, non-PMMA) material choice.

Section VI-C’s relabeling of “lateral chromatic spread” to “internal guided-angle spread” is a correction to what we call the quantity, not a fix to the specification vector: the record design’s period appears to have been selected substantially to minimize this internal-geometry term rather than a vali-

dated output-image quantity, and the metric that should replace it, most directly $d\theta_d/d\lambda$ evaluated per RGB primary and multiplied by a realistic LED linewidth (FWHM 20-25 nm), is closed-form and computable from the existing engine without new RCWA solves. At the record design we compute $d\theta_d/d\lambda \times \text{FWHM} \approx 2.3\text{-}2.9^\circ$ (blue), $2.8\text{-}3.5^\circ$ (green), $5.2\text{-}6.5^\circ$ (red) across that FWHM range, a genuine, non-cancelling angular blur estimate that is a natural replacement objective; retraining the tandem network and re-running the neural-adjoint search against it, which will very likely relocate the record period, has not been done in this revision. The field-edge transmission metric's redundancy with total transmission (Table 3's footnote) is likewise diagnosed, not yet fixed; making it independent requires the angle-resolved RCWA grid described below. Similarly, Section III-D shows that $\text{MTF}_{\text{chrom}}$'s $[0.543, 0.999]$ structural floor and its rapid oscillation in spatial frequency together let the record design's period reach a constructive fringe of Equation (6) at exactly 40 cyc/mm rather than a representative value of chromatic performance; replacing the three delta functions with finite-bandwidth LED line shapes, or integrating $\text{MTF}_{\text{chrom}}$ over a band of spatial frequencies rather than sampling one point, would remove both the floor and the fringe sensitivity, and neither is implemented here.

Refractive index is treated as a free design variable, not consistently as a material property: $n \in [1.48, 1.50]$ is fit to a single scalar per design and the search is free to place it anywhere in that range for all three RGB wavelengths simultaneously, but PMMA's actual dispersion gives distinct indices at each wavelength. Using the single-term Sellmeier fit for PMMA in the literature [23], $n^2 - 1 = 1.1819\lambda^2/(\lambda^2 - 0.011313)$ with λ in micrometers, we compute $n(450) = 1.5006$, $n(532) = 1.4937$, and $n(635) = 1.4886$, and the guided-window constraint that turns out to bind hardest is at the red wavelength, exactly where the record design's single scalar $n = 1.500$ overstates real PMMA's index by the largest margin. Using these three per-wavelength indices rather than one scalar $n = 1.500$, the record design's full-RGB guided window narrows from $[-0.24^\circ, 4.76^\circ]$ (5.00° wide) to $[-0.24^\circ, 4.11^\circ]$ (4.35° wide), a roughly 13% reduction we compute directly from the dispersion-corrected per-wavelength window inequality rather than approximate. Replacing the scalar index with a Sellmeier $n(\lambda)$ would also drop the design vector from eight parameters to seven; we have not yet made either change or re-run any search against it.

The roughness-MTF attenuation coefficient and the grating/coupler MTF weighting terms (Section III-D) remain literature-informed heuristics rather than closed-form derivations. Nilsen et al.'s directly relevant roughness-to-MTF measurements for PMMA and glass AR waveguides [12] are one reconciliation target; a more direct one, not yet used in this revision, is Kuang, Liu, and Shi's closed-form relationship between waveguide surface roughness and imaging quality at a stated spatial-frequency benchmark [29], which is closer in form to what the cascade's $\text{MTF}_{\text{rough}}$ term needs than an empirical curve is. Neither reconciliation is attempted term

by term here. That work is more urgent than it looked in an earlier draft: extending the existing $\text{MTF}_{\text{rough}}$ heuristic (Section III-D) from this paper's coated-surface σ bounds out to Nilsen's uncoated value ($\sigma = 15$ nm, Section III-C) collapses the term to numerically zero, which is not a physically credible prediction and instead shows that the heuristic's coefficients were only ever exercised, and implicitly tuned, within the narrow coated-roughness range this paper uses. The transmission-side roughness treatment (Section III-C's Tien/Debye-Waller formula) does not have this problem and gives the physically reasonable 30%-loss estimate reported there; the MTF-side heuristic needs recalibration against Nilsen's measured MTF curves, not merely a wider input domain, before it can be trusted outside the coated regime.

The 90,090-point RCWA calibration grid (Section IV-B) covers only normal incidence and a binary (rectangular) groove profile; angle-dependent and non-binary (blazed, slanted, sinusoidal) profile support exists in the rigorous-solver layer (Section IV) but is not yet folded into the coupling term the training loop and design searches actually use, and that gap is the direct reason $A(\theta_i)$ (Section III-C) and the field-edge metric are not RCWA-anchored. Extending the grid with a θ_i axis (roughly 7-9 points spanning $\pm 6^\circ$) and rebalancing the existing axes toward the period resolution Section IV-B identifies as the current bottleneck, at the cost of coarsening the over-resolved depth axis, are the concrete next steps. As one check that does not require a grid rebuild, we swept the depth-optimal coupling efficiency $\arg \max_d \eta_1(\Lambda, D)$ across the full existing 5×13 (period, duty) grid at $n = 1.500$ and 532 nm, separately for each polarization and for their unpolarized average, and the result is more informative than a flat confirmation would have been: for TE, all 65 (period, duty) cells favor a depth between 195 and 240 nm, a tight, robust interior optimum across the entire guided-window plane. For TM, the optimum is not robust at all: only 43% of cells favor a depth below 250 nm, the rest favor a depth above 350 nm (up to the scalar model's own 400 nm search boundary), and the unpolarized average, which blends the two, splits roughly down the middle (54% below 250 nm). Read together with Section VI-F's finding that the record design is 96.9% TE, this sharpens rather than weakens the paper's central depth-optimum claim: the interior optimum near 200 nm is a robust, plane-wide property of the polarization channel this device actually uses, and the unpolarized average that Table 2 and Section VI-C otherwise lead with obscures that robustness by mixing in a polarization whose optimum behaves altogether differently. A fully TE-scoped restatement of the depth-optimum claim, in place of the unpolarized framing used throughout Sections IV and VI, is a natural next revision. Separately, we checked whether the record design's own trilinear interpolant is accurate specifically along the index axis, the axis Section IV-B flags as steepest and least sampled: solving three off-node index points ($n = 1.485, 1.495, 1.4975$) at the record design's exact period, depth, and duty and comparing against the interpolant gives absolute errors of 4.8×10^{-4} - 4.9×10^{-4}

(TE) and 2.1×10^{-4} - 2.3×10^{-4} (TM), essentially the same magnitude as the interpolant’s error at the index axis’s own grid nodes evaluated at this same off-grid period, depth, and duty (4.8×10^{-4} TE, 2.0 - 2.4×10^{-4} TM); the index axis is not contributing meaningfully more error than the other three axes already do, contrary to what its coarser, three-point sampling might suggest. A dedicated substrate-side (out-coupler-geometry) rigorous solve to check the reciprocity assumption in Equation (3) directly, rather than argue for it, is the single cheapest item on this list and was not run in this revision because we were not confident of getting the incident-medium convention right under this revision’s time constraints without risking reporting a wrong “verified” number, which seemed worse than reporting the assumption as open.

Every rigorous number in this paper comes from one implementation (grcwa). RCWA convergence was checked at the scalar-optimal geometry only (Section IV-A), and TM convergence for one-dimensional gratings is generally more delicate than TE; cross-validating roughly ten points, weighted toward TM, against an independent solver such as *torcwa* [25], *S4*, or a full-wave FDTD code would be cheap insurance for the polarization-resolved claims this paper leans on most (Section VI-F) and has not been done. Section III-C documents the shipped engine’s factor-of-two error in N_b and ℓ , quantifies its (negligible) numerical impact on every result in this paper, and explains why we corrected the paper’s equations without patching the code this cycle: the physics-probe checkpoint system (Section V-A) would correctly invalidate every existing checkpoint the moment the live formula changed, and a full retrain was outside this revision’s scope. The code fix, and the retrain it forces, is the first thing that should happen before any new design search, and that retrain is also the natural time to widen σ ’s upper bound to Nilsen’s uncoated value and let a fresh search find whatever design, if any, partially compensates for the higher roughness (Section III-C) rather than only evaluating the existing coated-optimal geometry under the uncoated hypothesis as we do here.

A partial, verified sensitivity check is reported in this revision rather than a formal one. $S(L_c)$ (Section III-C) varies by $\sim 30\%$ over the PMMA design bounds, not negligibly, and that variation matters more than it first appears: at Nilsen’s uncoated $\sigma = 15$ nm, the same L_c range moves roughness-scattering transmission by roughly nine percentage points (from 61% to 70% transmitted) rather than the fraction of a percentage point it moves at this paper’s coated σ bounds, so correlation length is a genuinely load-bearing parameter for an uncoated device even though it is nearly inert for a coated one. Bulk transmission $\exp(-\alpha\ell)$ varies from 99.87% to 98.74% across the PMMA-mode α bounds (5×10^{-5} to 5×10^{-4} mm^{-1}), and roughness transmission at this paper’s coated σ bounds (0.7-1.1 nm) varies from 99.98% to 99.81%: both channels stay in a collectively low-loss regime across their respective PMMA bounds, so essentially all of the transmission-optimization opportunity in this design space concentrates in the grating-coupling term, exactly the term Section IV recalibrates, for the coated device

this paper’s record design represents. This one-parameter-at-a-time, bound-range check is not a substitute for a proper variance-based (Sobol) decomposition across all eight parameters and both output metrics, which would also resolve how many of the eight parameters the R^2 values of Section VI-A are effectively fit against; that full analysis has not been run.

Exit-pupil expansion and eyebox uniformity, a binary (unslanted) grating profile, world-side ghost orders and see-through distortion, and non-normal-incidence coupling physics are outside what this paper’s engine models at all; this is stated plainly in Section III-A rather than left for a reader to discover, and Lee, Lee, and Chung’s adjoint-designed couplers [9] are the closest treatment of several of these we are aware of. As with essentially all inverse-design work at this stage, the reported designs are not checked against manufacturing tolerances, e.g. the achievable duty-cycle and depth precision in nanoimprint or e-beam fabrication of a 200 nm-deep, sub-450 nm-period grating in PMMA; a tolerance sweep around the record design (reporting how much the specification vector degrades under a plausible $\pm\sigma_d, \pm\sigma_D$ fabrication error) is a short, concrete next step that has not been run. Finally, the transmission cascade of Equation (3) evaluates the grating-coupling term, the guiding mask, and the diffraction angle at the green design wavelength only for every reported transmission metric; a fully wavelength-resolved transmission treatment (as already exists for the guided-angle-spread metric) is a natural extension and would also remove the specific green-only blind spot noted in Section VI-C, where red’s marginal guiding status at the field-edge angle does not currently affect the reported field-edge transmission number at all.

None of the above were treated as complete anywhere in this paper; where a result depends on one of them, the text says so explicitly rather than letting the gap surface only in this section.

IX. CONCLUSION

The central, load-bearing finding of this paper is that the guided-mode condition for a full-color, low-index polymer waveguide forces its grating period below every design wavelength, exactly the sub-wavelength regime in which scalar diffraction theory is known to fail, and that the failure is not a rounding error: it overestimates coupling efficiency by a factor of 5 to $16\times$ at the design our own scalar-driven search first converged to, and correcting it, by calibrating the coupling term against 90,090 rigorous electromagnetic solutions, moves the true efficiency optimum off the scalar model’s search-boundary artifact and onto a rigorously verified interior optimum near 200 nm depth. Built around that finding, we presented a physics-anchored tandem neural inverse-design framework for PMMA diffractive AR waveguides, with an exact differentiable analytic engine as the tandem’s decoder rather than a second learned surrogate, and an explicit accounting of what that engine does and does not model (Section III-A). The forward surrogate, tandem inverse network, and neural-adjoint search each pass a reproducible honesty check (a physics-probe checkpoint-invalidation sys-

tem, a four-part memorization audit, and independent rigorous re-verification of every headline design) rather than being reported on validation-set performance alone, and we extend that same discipline to the paper’s own specification vector: one of the four system metrics was mislabeled at first draft (an internal guided-angle spread reported as an output chromatic aberration), a second carries no information independent of total transmission, and a material roughness bound was traced to the wrong process condition in the source literature (a coated surface’s measured value, applied to a design described as coating-free); each is corrected here with the closed-form or re-derived number that replaces it, and Sections III-C and VIII record where a first-pass correction was itself imprecise once checked a second time. We hope this iterated, self-adversarial verification discipline, as much as the specific numbers, is useful to the next waveguide inverse-design paper that calibrates a fast training-time model against a slow, correct one, and to the next paper that checks its own specification vector against the literature it cites.

DATA AVAILABILITY STATEMENT

The differentiable physics engine, network training scripts, trained checkpoints, the RCWA calibration grid and its off-grid audit, the verification scripts behind every corrected number in this revision, and every results table cited in this paper are maintained in the author’s project repository and are self-contained; the engine is validated directly in this paper (Section III) against literature closed forms and rigorous electromagnetics rather than against an external, unpublished companion document. Data and code will be made available at the time of formal submission or upon reasonable request.

ETHICS DECLARATION

This work involved no human subjects, animal subjects, or personally identifiable data. No ethics review was required.

AUTHOR CONTRIBUTIONS

Connor Wang: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing (original draft and review & editing), visualization.

CONFLICT OF INTEREST

The author declares no conflict of interest.

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AI-USAGE DISCLOSURE

A large language model (Claude, Anthropic) was used, under the author’s direction, to assist with code development, with independent numerical verification of results reported in Section VI against the project’s saved network checkpoints and physics engine, and with drafting and editing this manuscript’s text. All modeling decisions, all reported results, and the final text were reviewed and are the responsibility of the author.

NUMERICAL VERIFICATION NOTE

The per-metric tandem and baseline errors in Table 3, and every corrected number introduced in this revision, were independently computed by the author using a from-scratch NumPy port of the physics engine and, where relevant, the trained network’s forward pass, rather than taken from a training run’s printed summary. The NumPy physics port was validated by reproducing the project’s on-file record design (Section VI-C) to five significant figures across all four specification metrics before being used for evaluation.

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